Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

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Abstract

Falls, frailty, and chronic conditions such as cardiovascular disease, diabetes, and obesity remain leading causes of morbidity and mortality in older adults. Yet, existing predictive models often rely narrowly on physical health indicators, overlooking nutrition and lifestyle factors that may enhance early detection of risk. This thesis develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in a cohort of older adults. The models integrate physical, mental, lifestyle, and socio-demographic predictors, with particular emphasis on under-explored nutritional and behavioural variables highlighted in prior research. Multiple approaches are compared, including logistic regression, random forests, and gradient boosting, with performance assessed via cross-validation and feature importance analysis. By identifying high-impact, modifiable predictors, this work aims to advance early risk stratification, guide targeted prevention strategies, and inform public health policy to support healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this paper without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This paper has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor and Dr. Name Surname at University of Limerick.

Limerick, 2026

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Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AUC Area Under the ROC Curve

CES-D Center for Epidemiologic Studies Depression Scale
CV Cross-Validation

DNN Deep Neural Network

F1 F1-score (harmonic mean of precision and recall)
FI Frailty Index

GBM Gradient Boosting Machine
GDS Geriatric Depression Scale

HRS Health and Retirement Study, (a longitudinal study of ageing in the United States)

LDA Linear Discriminant Analysis

ML Machine Learning
MPOD Macular pigment optical density

Recall Recall (true positive rate)
RF Random Forest

SHAP Shapley Additive Explanations
SHARE Survey of Health, Ageing and Retirement in Europe

TILDA The Irish Longitudinal Study on Ageing
TUG Timed Up and Go (mobility test)

List of Abbreviations

WHO	World Health Organization

Chapter 1

Introduction

1.1 Background and Context

Chronic conditions, including cardiovascular disease, diabetes, and frailty, are leading contributors to morbidity and mortality in ageing populations [1, 2]. These conditions are associated with declines in functional capacity (e.g., gait speed, grip strength [3]), increased healthcare utilisation [4], and reduced quality of life [5]. Traditional risk prediction models rely primarily on clinical biomarkers such as blood pressure and lipid profiles. However, increasing evidence highlights the importance of multidimensional predictors, including nutrition [6], physical activity [7], mental health [8], and social determinants [9].

The Irish Longitudinal Study on Ageing (TILDA) provides a uniquely rich, nationally representative dataset of adults in Ireland aged 50+ with repeated measures spanning clinical, nutritional, lifestyle, cognitive, and psychosocial domains. Leveraging TILDA waves 1–5[1], the harmonised longitudinal dataset, and the IDS dataset (covering participants with disabilities), this project has access to comprehensive repeated measures across multiple domains. Previous studies using TILDA have linked, for example, slower blood pressure recovery after standing to accelerated brain ageing [10], and vitamin D deficiency to incident prediabetes [11]. Such findings highlight the need for integrated models that combine traditional and non-traditional risk factors.

[1] TILDA data are pseudonymised and accessed under licence via the Irish Social Science Data Archive (ISSDA); use adheres to UL S&E ethics approval and data governance requirements.

Leveraging TILDA waves 1–5, the harmonised longitudinal TILDA dataset, and the IDS dataset (covering participants with disabilities) offers a comprehensive set of repeated measures across multiple domains, enabling robust analysis of multidimensional predictors.

1.2 Research Problem and Motivation

Current Machine Learning (ML) approaches for chronic disease prediction in older adults often fall short in several ways.

1.2.1 Limited Integration of Multidimensional Data

Most models prioritise clinical variables but neglect modifiable lifestyle and nutrition factors [6, 12].

1.2.2 Underuse of Longitudinal Dynamics

Few studies exploit repeated measures to capture trajectories of risk, such as frailty progression [13].

1.2.3 Lack of Explainability

Many ML models are not interpretable, limiting clinical utility [14].

Addressing these gaps requires methods that integrate multidomain predictors, account for longitudinal dynamics, and balance predictive accuracy with interpretability.

1.3 Research Aims and Objectives

This thesis aims to develop and validate ML models that integrate nutrition, lifestyle, and mental health features with clinical data to improve prediction of chronic disease and ageing-related outcomes in a representative cohort of older adults.

Specific objectives are:

- Conducting a focused literature review on ML for chronic disease prediction in ageing populations, emphasising gaps in lifestyle and nutrition integration.
- Preprocess and harmonise TILDA waves 1–5, engineering multidomain features including nutrition, activity, mobility, mental health, and clinical indicators.
- Develop and compare baseline statistical models (e.g., logistic regression) with advanced ML methods (e.g., Random Forest (RF), Gradient Boosting Machine (GBM)).
- Evaluate models using discrimination (Area Under the ROC Curve (AUC)), calibration, and error metrics (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)); conduct ablation analyses to quantify the added value of lifestyle and nutrition.
- Identify actionable predictors (e.g., vitamin D status, gait speed reserve) and highlight modifiable risk factors relevant to intervention.
- To Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.4 Methodology Overview

The methodological workflow consists of the following components.

1.4.1 Data Curation

1.4.1.1 Harmonisation

This project uses the official harmonised TILDA dataset, which consolidates variables from waves 1–5 to ensure consistency across repeated measures and comparability with other international longitudinal studies such as the Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS) and Survey of Health, Ageing and Retirement in Europe (SHARE). Where definitions differ between waves (e.g., frailty indices, nutritional measures), harmonisation rules are documented in the official dataset.

1.4.1.2 Handling Missing Data

Missingness will be addressed using both complete-case analysis (sensitivity check) and multiple imputation methods, depending on the type and extent of missing data. This ensures robustness of findings and reduces bias introduced by attrition across waves.

1.4.1.3 Feature Engineering

Derived variables will be constructed to capture multidimensional predictors. Examples include:

- Frailty indices (e.g., Fried's phenotype, Frailty Index (FI)).
- Mobility measures (e.g., Timed Up and Go (mobility test) (TUG), gait speed, grip strength).
- Mental health scales (e.g., Center for Epidemiologic Studies Depression Scale (CES-D), Geriatric Depression Scale (GDS)).
- Nutrition scores (e.g., dietary diversity indices, micronutrient biomarkers).

1.4.2 Modelling

Development of baseline (logistic regression) and advanced ML models (RF, GBM) tuned via Cross-Validation (CV).

1.4.3 Evaluation

Assessment with discrimination metrics (AUC), calibration (plots, Brier score), and error measures (accuracy, Recall, F1). Explainability will be examined using feature importance and Shapley Additive Explanations (SHAP) values.

1.5 Contributions of the Research

This research makes contributions at three levels:

- **Empirical:** Demonstrates the added predictive value of nutrition and lifestyle features beyond clinical baselines in chronic disease risk prediction.
- **Methodological:** Provides a reproducible pipeline for harmonising longitudinal TILDA data and integrating multidomain features into ML models.
- **Clinical:** Highlights modifiable predictors with potential for targeted interventions and policy development to promote healthy ageing.

1.6 Thesis Outline

The remaining chapters of this dissertation are structured as follows:

Chapter Two presents a focused literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle and nutrition predictors and methodological considerations. *Chapter Three* describes the TILDA study design and data, the preprocessing pipeline, harmonisation decisions, and feature engineering strategy. *Chapter Four* outlines the modelling framework, including baseline statistical models and advanced ML, hyperparameter tuning, validation, and evaluation metrics. *Chapter Five* reports results, compares model performance, and presents feature importance and ablation analyses. *Chapter Six* discusses implications, limitations, ethical and governance considerations, and opportunities for future work.

A series of documents have been included in the Appendix section of this dissertation. These are:

- *Appendix A*: Variable codebook and wave harmonisation notes (including anonymisation actions, top-coding, and grouping rules relevant to the features used).
- *Appendix B*: Ethics.

- *Appendix C*: Modelling details, additional tables/figures, and reproducibility checklist (overview of data processing scripts).

A digital archive is attached to this dissertation containing:(Access request required)

- *folder 1*: Preprocessing and feature-engineering scripts (documentation only; no raw TILDA data).
- *folder 2*: Model training/evaluation code and configuration files to reproduce reported results.

Chapter 2

Literature Review

2.1 Introduction

- Purpose of this chapter
- Brief overview on TILDA as a unique dataset (rich, longitudinal, multi-domai...)
- Scope: frailty, mental health, chronic disease, multimorbidity, nutrition, and ML models

2.2 Chronic Disease and Ageing

- The burden of chronic conditions in older adults
- Healthy ageing framework World Health Organization (WHO)
- Importance of multimorbidity, frailty, mental health

2.2.1 Frailty and Physical Function

- Gait speed, grip strength, functional mobility decline
- Predictors of frailty transitions ([15], [16], [17])

2.2.2 Mental Health in Older Adults

- Depression, anxiety, loneliness, social connectedness ([18], [19], [20])
- Links between physical activity and depression ([8], [7], [12])

2.2.3 Nutrition and Biological Ageing

- Vitamin D, lutein, zeaxanthin, Macular pigment optical density (MPOD) ([6], [11], lairdVitaminDeficiencyIreland2020, feeneyLowMacularPigment2013)
- Malnutrition and its predictors ([2])
- Biological vs chronological ageing ([21], [22])

2.3 Machine Learning in Ageing and Chronic Disease

- Overview: Why ML is valuable in ageing research [23]
- Transition from regression-based models → ensemble ML → deep learning(cite a example here)

2.3.1 Feature Sets in ML Studies

- Clinical, sociodemographic, lifestyle, cognition, psychosocial
- Over-reliance on physical health predictors vs underuse of nutrition/social variables

2.3.2 ML Models Used

- Logistic regression, Markov models, Linear Discriminant Analysis (LDA) (traditional)
- Ensembles RF, GBM, Deep Neural Network (DNN) – e.g., ([24]) diabetes prediction

- Unsupervised clustering for dementia ([25])
- Brain age prediction models ([26], [10])

2.3.3 Performance and Validation

- AUC, R^2 ranges across tasks (diabetes, gait speed, mortality, depression)
- Limitations: internal validation only, cross-sectional data, no generalisation

2.3.4 Explainability in ML for Ageing

- SHAP, TreeExplainer examples ([24], [27], [3])
- Challenges of black-box models in clinical practice

2.4 Gaps in the Literature

- Limited population diversity (mostly Irish, 50+)
- Over-reliance on physical predictors; neglect of nutrition, social, religious/spiritual, environmental variables
- Dominance of regression models; limited advanced ML such as neural networks, transformers
- Validation gaps – very few external validations, especially for ensemble and brain-age models
- Underexplored outcomes: pain, sleep, QoL
- Cross-sectional bias; weak time-series modelling
- Poor transparency on missing data handling
- Limited integration of nutrition biomarkers
- Lack of biological age prediction vs chronological
- Interpretability challenges in ML

2.5 Chapter Summary

- I need to recap the main findings from LR systematic mapping excel file
- Identify your contribution niche:
  - integrating nutrition + lifestyle + clinical predictors
  - ML + explainability
  - time-based prediction (longitudinal TILDA data)
- Lead into Methodology (Chapter 3)

Chapter 3

Data and Preprocessing

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).

Figure 3.1: Title of Figure - Description. [Source: (**?**)]

3.2 Creating a list

- **(2000)** item 1.
- **(2004)** item 2.
- **(2010)** item 3.
- **(2013)** item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

> Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.
(**?** , verses 208-218)

Chapter 4

Methodology

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol ^{T} denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.5}$$

$$= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{\mid A_{ba} \mid\mid M_{bm} \mid \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}}$$

$$= \frac{-0.2338, -0.0931, -0.0651}{0.2601}$$

$$= (-0.8989, -0.3579, -0.2503)$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

Chapter 5

Results

5.1 References

Open the file 'references_myphd' with BibTex. The file is located in '/7_references/'

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book**: (**?** , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter**: (**?**)
- **Book Contribution**: (**?**)
- **MUSIC CD/DVD**: (**?**)
- **Journal Article**: (**?**)
- **Patent**: (**?**)
- **PhD/Master Dissertation**: (**?**)
- **Conference Paper**: (**?**)
- **Technical Report**: (**?**)

- **Unpublished Report**: (**?**)
- **Web Videos**(**?**) (. . . not the name of the person who uploaded the video though)
- **Web Paper**: (**?**)
- **Web Journal Article**: (**?**)
- **Webpage**: (**?**)
- **Downloaded Images or Sounds**: (**?**)

You can also you this kind of referencing if needed in the text:
Surname [**?**] said that . . .

Chapter 6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

Appendix A

Insert a figure

Appendix

Appendix B

Title

Type here you text as usual . . .

Appendix

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 *  Copyright (c) 2013, Giuseppe Torre
 *  All rights reserved.
 *
 *  Redistribution and use in source and binary forms, with or without
 *  modification, are permitted provided that the following conditions are met:
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 *  ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 *  (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 *  SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 *
 */



#include "ext.h"
#include "ext_mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//******************CALIBRATION STUFF*************************//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX_OFFSET 2043 //initial offset in accelerometer X
#define ACCELY_OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ_OFFSET 2264//initial offset in accelerometer Z

#define MagX_OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```c
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667//initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
//****************** END CALIBRATION STUFF ************************//


void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
        t_object m_ob;    // Must always be the first field; used by Max

        Atom m_args[9];   // we want our inlet to be receiving a list of 10 elements

        long m_value;     //inlet

        void *m_R1;             //these are all the outlets for the 3 X 3 Matrix
        void *m_R2;             // R1 --> firts top left cell ..R2 middle cell of the first
            row ...and so on
        void *m_R3;                                   //
        void *m_R4;                                   // End Outlets
            ________________________________________
} t_triad;

                        void *triad_new(long value);


                        void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
                            s);
                        void triad_free(t_triad *triad);



                        void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);




                        void MatrixByMatrix(double *Result,double *MatrixLeft,double *
                            MatrixRight);

                        void Matrix2Quat(double *Quat,double *Matrix);
                        void Quat2Matrix(double *Matrix, double *Quat);
                        void inverseQuat(double *InvQuat, double *RegQuat);
                        void NormQuat(double *YesQuat, double *NotQuat);
                        void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
                        void NormVect(double *YesVect, double *NotVect);
                        double orientationMatrix[9];
                        double Result[9];
                        int     i;
                        double InvorientationMatrix[9];

                        double temp[6];
                        double ref[6];
                        double vectAx[3];
                        double vectAy[3];
                        double vectAz[3];
                        double vectBx[3];
                        double vectBy[3];
                        double vectBz[3];
                        double MagnCrosProd_A;
                        double MagnCrosProd_B;
                        double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
                            VectBynorm, VectBznorm;
```

```
                            double m[9], n[9];
                            double quat_e[4];
                            double invquat_e[4];
                            double mult;
                            double quat_new[4];
                            double quat_old[4];
                            double ecs,accex_ADCnumber,y,accey_ADCnumber,z,accez_ADCnumber,mx,
                                 magnx_ADCnumber,my,magny_ADCnumber,mz,magnz_ADCnumber;
// SLERP Variables
                            double trace, Suca;
                            double tol[4], omega, sinom, cosom, scale0, scale1, tez,
                                 orientationMatrixA[9];

int main(void)
{
        // set up our class: create a class definition
        setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
             sizeof(t_triad), 0L,A_GIMME, 0);
    addmess((method)triad_list, "list", A_GIMME, 0);
        addmess((method)triad_assist, "assist", A_CANT, 0);
        finder_addclass("Maths","triad");
        post("....I'm-TRIAD-Object!....from-AHRS-Library...",0);
        return 0;
}

/* -------------triad_new ---------------*/

void *triad_new(long value)
{
        t_triad *triad;
        triad = (t_triad *)newobject(this_class);        // create the new instance and return
              a pointer to it
        triad->m_R4 = floatout(triad);
        triad->m_R3 = floatout(triad);
        triad->m_R2 = floatout(triad);
        triad->m_R1 = floatout(triad);

return(triad);
}

etc.

etc.
```

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